

CHAPTER 1

INTRODUCTION

Gas leakage is one of the most dangerous hazards that can occur in both residential and industrial areas. With the increase in the use of LPG and other combustible gases, it has become essential to ensure proper safety measures are in place. The traditional methods of gas detection often lack automation and quick response, which can lead to fatal consequences. This project introduces an intelligent, IoT-based solution to address these challenges effectively.

1.1 INTRODUCTION OF THE PROJECT

In today's world, the use of Liquefied Petroleum Gas (LPG) and other combustible gases is extremely common in households, hotels, laboratories, and industries. These gases play a crucial role in daily activities, particularly for cooking and heating purposes. However, their highly flammable nature makes them dangerous, especially when leaks go undetected. Gas leaks can result in fires, explosions, property damage, and serious threats to human life. Unfortunately, many incidents occur due to a lack of timely detection and response.

This project introduces an IoT-based Gas Leakage Detection System with an Automatic Regulator On/Off mechanism to mitigate such risks effectively. The system is designed to detect the presence of gases like LPG and methane in the environment and take immediate corrective action without human intervention. The key goal is to enhance safety by combining smart sensor technology with real-time automated responses.

At the heart of this system is the MQ-5 gas sensor, which is capable of detecting various combustible gases, including LPG, methane, and hydrogen. When a gas leak is detected, the sensor sends signals to the Arduino Uno microcontroller, which acts as the brain of the system.

The Arduino processes the data and triggers multiple safety mechanisms to prevent any hazardous outcome.

One of the primary actions taken by the system is the automatic shutdown of the gas supply. This is achieved through the use of a servo motor that is mechanically connected to the gas regulator. Upon detection of a gas leak, the servo motor is activated to close the regulator, thereby stopping the flow of gas and minimizing the risk of further leakage or ignition.

In addition to cutting off the gas supply, the system activates an exhaust fan connected via a relay module. The fan helps to ventilate the area by dispersing the leaked gas, reducing its concentration in the air and making the environment safer. This additional action ensures that the gas does not accumulate to explosive levels.

The system also includes a buzzer to provide an audible alert to anyone nearby. This is crucial in ensuring that even people without access to the visual indicators can be warned in time. Along with the buzzer, the system uses LED indicators and an LCD display for visual feedback. The LCD, connected via an I2C module, shows real-time status updates such as "Gas Detected" or "Safe Environment."

The red and green LEDs act as simple, immediate indicators of system status. When the environment is safe, the red LED remains on. If gas is detected, the green LED turns on to alert the user. This makes the system easy to interpret, even for individuals without technical knowledge.

One of the innovative aspects of this system is its potential for IoT integration. Although the initial implementation may not include GSM or Wi-Fi modules, the design supports their future addition. With GSM, the system can send SMS alerts to the user, while Wi-Fi connectivity

enables mobile app notifications and remote monitoring. These features align the project with current smart home safety trends.

The modular and scalable nature of this system allows it to be adapted for various use cases. Whether in residential homes, commercial kitchens, or small industrial units, the system can be tailored to meet specific safety requirements. Additional sensors can also be incorporated to detect other hazardous conditions like fire or temperature spikes.

Energy efficiency is another highlight of the system. All components are selected with low power consumption in mind, ensuring continuous operation without excessive energy use. A backup power supply can also be included to maintain functionality during power outages, making the system reliable even in emergency situations.

From an engineering perspective, this project is a comprehensive blend of embedded systems and IoT. It offers students hands-on experience with real-time data processing, automation, sensor integration, and microcontroller programming. This makes it both an educational and practical initiative.

The integration of both hardware and software plays a critical role in achieving the system's objectives. The Arduino Uno serves as the brain of the project, interpreting sensor data and executing actions based on predefined conditions in the code. The logic is simple yet effective, allowing quick detection and timely activation of safety measures. This fusion of embedded systems and IoT principles makes the project both innovative and educational for engineering students.

The system contributes to raising public awareness about the importance of automated safety mechanisms. It demonstrates how accessible technology can be used to safeguard lives and property, encouraging the adoption of smart solutions in everyday settings.

Moreover, the project showcases the practical application of core engineering principles. By translating theoretical knowledge into a functioning safety system, it bridges the gap between classroom learning and real-world problem-solving. It is an excellent example of how engineering projects can have a direct, positive impact on society.

In conclusion, the IoT-based Gas Leakage Detection System with an Automatic Regulator On/Off mechanism offers an efficient, scalable, and cost-effective solution to a common yet serious safety challenge. Through automation, real-time monitoring, and user-friendly features, the system ensures a safer environment and reflects the potential of smart technology in improving quality of life.

1.2 PURPOSE OF THE PROJECT

The primary purpose of this project is to design and implement an intelligent and responsive IoT-based gas leakage detection system with an automatic regulator on/off mechanism. This system is developed to enhance safety in residential, commercial, and small-scale industrial environments by enabling real-time gas leak detection and automated preventive actions without human intervention.

Gas leaks pose significant risks, including fire hazards, explosions, suffocation, and long-term health issues. Traditional systems often rely on manual detection or basic alarms, which may not be sufficient to prevent catastrophic outcomes. This project aims to overcome such limitations by offering a solution that not only detects the presence of harmful gases like LPG and methane but also takes immediate corrective action to prevent further escalation.

A key purpose of this project is to ensure **early detection** of gas leaks. By continuously monitoring the environment using the MQ-5 gas sensor, the system can detect even minor leaks

before they become dangerous. This rapid response time reduces the chance of accidents, safeguarding human life and property.

Another important goal is to enable **automated intervention**. Once a gas leak is identified, the system automatically turns off the gas regulator using a servo motor, preventing additional gas from escaping. This automation removes the dependence on human response, which can often be delayed, especially in unattended areas.

The project also serves to improve **local and remote awareness** of hazardous conditions. Through the use of an LCD display and LED indicators, individuals in the vicinity are informed of the system's status in real time. The buzzer provides an immediate audible alert to notify nearby users of a potential danger.

Moreover, this system is designed with **future integration in mind**. The addition of GSM and Wi-Fi modules enables real-time notifications to be sent to users' mobile devices. This ensures that even if users are not physically present, they can still monitor and respond to gas leaks from remote locations, improving responsiveness and control.

The system is built to be **user-friendly and cost-effective**, making it accessible for homes and small businesses. By using commonly available components such as Arduino Uno, MQ-5 sensor, and relay modules, the design remains affordable while delivering reliable performance. This makes the system suitable for widespread adoption.

The project also promotes **energy efficiency**. As it operates continuously, low-power components are selected to ensure that the system runs 24/7 without significantly increasing electricity consumption. This makes it suitable for long-term deployment in both domestic and industrial settings.

A crucial educational purpose of the project is to give engineering students **hands-on experience with embedded systems and IoT** technologies. By working on this project, students gain practical knowledge of sensors, actuators, microcontrollers, and real-time systems—skills that are highly valuable in today’s tech-driven world.

In addition to detecting LPG, the MQ-5 sensor can identify other combustible gases like methane and hydrogen. This expands the utility of the system beyond kitchens to laboratories, fuel stations, and storage facilities. Therefore, another purpose is to **offer a multi-purpose safety system** that can be deployed in diverse environments.

This project is intended to raise **awareness about gas safety** and demonstrate how technology can be used to address everyday risks. By integrating IoT with automation, the system provides a smart solution that reduces human dependency and increases responsiveness in emergency situations.

From a social perspective, the project aims to **enhance public safety**. In many regions, gas-related incidents are common due to poor infrastructure or lack of monitoring. Deploying such systems can significantly reduce the frequency and severity of such accidents, making homes and workplaces safer.

The purpose of this project also includes promoting **technological innovation in safety systems**. By combining software and hardware in a cohesive system, the project showcases how IoT and automation can be leveraged to create smart environments that respond proactively to risks.

Scalability is another key focus. The system is designed in a modular fashion, allowing for **easy upgrades and customization**. For example, more sensors can be added to increase

coverage, or cloud integration can be introduced for real-time analytics. This makes the system adaptable to evolving needs and technologies.

In terms of reliability, the system includes features such as a **backup power supply** to ensure continuous operation even during power failures. This adds robustness and increases confidence in the system's ability to function during critical moments.

Lastly, the overarching purpose of this project is to **bridge the gap between academic learning and practical application**. It encourages students to develop real-world solutions that address pressing issues using the latest technology trends. This not only benefits the learners but also contributes to the community by promoting smarter and safer living environments.

CHAPTER 2

SYSTEM ANALYSIS

The system analysis identifies the limitations of existing gas leakage detection systems, which rely on manual intervention and lack automation and remote monitoring. The proposed IoT-based system addresses these issues by integrating real-time gas detection, automatic gas shutoff, and remote monitoring via GSM or Wi-Fi. It aims to enhance safety through faster response times and automated actions, such as activating exhaust fans and alerts. The system is designed to be scalable and adaptable, ensuring it can incorporate future technological advancements.

2.1 EXISTING SYSTEM

In today's safety landscape, traditional gas leakage detection systems are commonly deployed in households, industries, and laboratories to identify and alert users about gas leaks. These systems are usually basic in functionality, relying on simple gas sensors and buzzers to notify nearby individuals when leakage occurs.

The most common type of existing system involves standalone gas detectors. These devices are mounted on walls or near gas sources and are equipped with sensors like MQ-5 or MQ-2. When gas is detected above a certain threshold, the device activates a buzzer or LED indicator to alert people nearby.

While these basic systems can detect leaks and raise alarms, they are limited by their inability to perform any corrective or preventive actions. This means that although users may be alerted,

they are still required to take manual actions, such as shutting off the gas regulator, ventilating the area, or calling for help.

These existing systems are non-intelligent and reactive rather than proactive. Their response is confined to sounding an alarm and expecting human intervention. In critical situations where every second matters, this delay in response can lead to severe outcomes, including fires, explosions, or gas poisoning.

Another major drawback of current systems is their lack of integration with modern technology. Most do not support communication modules such as GSM or Wi-Fi, which would allow users to receive alerts remotely. If a gas leak occurs while no one is present, the system's alarm may go unnoticed.

The absence of automated control features in traditional systems poses a significant risk. For instance, without an automatic shut-off mechanism, the flow of gas continues even after the leak has been detected, compounding the danger. This creates a reliance on the user being both present and responsive.

Furthermore, the user interface of existing systems is minimal or non-existent. Many systems lack display modules to show real-time status or gas concentration levels, limiting the user's ability to monitor and understand the situation clearly.

Visual indicators such as LEDs are often used, but without proper context or labeling, users may not fully grasp the meaning of the signal. This reduces the effectiveness of the alert system, especially in high-noise environments or during power outages where audible alarms might not be enough.

Maintenance and reliability are also concerns with older systems. Since most are not designed to be modular or upgradable, any component failure usually requires replacing the entire unit. Additionally, sensors degrade over time, and traditional systems do not include diagnostics to indicate sensor health or accuracy.

Cost can also be a limiting factor. Some existing systems, especially commercial-grade ones, are priced high, making them unaffordable for many households or small businesses. As a result, cost-effective alternatives are sought—but these often compromise on features and safety.

Portability and scalability are not priorities in existing systems. Most models are designed for fixed installation and cannot easily be moved or reconfigured. This rigidity makes them unsuitable for users who require more flexible safety solutions across multiple environments.

Power dependency is another issue. Many traditional systems do not have a backup power source. In the event of a power outage, the system becomes inoperative, leaving the space vulnerable to undetected gas leaks at a crucial time.

In industrial applications, more advanced systems may be used, such as those integrated into SCADA systems. However, these solutions are expensive, complex, and not suitable for small-scale or domestic use. Their installation and configuration require expert knowledge and maintenance contracts.

Additionally, most existing systems are not eco-friendly. Continuous high-power consumption or use of non-replaceable components contributes to increased energy usage and electronic waste, which goes against the modern push toward sustainable technologies.

In conclusion, while existing gas leakage detection systems have contributed to increased safety awareness, they fall short in providing a comprehensive and automated response. Their limitations in automation, remote alerting, modularity, and user-friendliness create a strong need for a more intelligent and proactive solution, such as the IoT-based gas leakage detection system proposed in this project.

2.2 PROPOSED SYSTEM

The proposed system aims to offer a smart, automated, and efficient solution to gas leakage hazards using an IoT-based embedded design. It is built with a strong focus on safety, affordability, ease of use, and real-time responsiveness. The core concept revolves around automatic detection of LPG gas leaks and immediate corrective action without human intervention.

At the heart of the system is the **Arduino Uno microcontroller**, which serves as the brain of the operation. It is programmed to continuously monitor inputs from the gas sensor and make decisions based on the readings. Arduino provides a reliable and cost-effective control platform, widely used in embedded systems and prototyping.

The **MQ-5 gas sensor** is a key component responsible for detecting the presence of LPG or other combustible gases like methane. This sensor operates by measuring gas concentration in the air and converting it into an analog signal that is sent to the Arduino for processing.

Once the gas concentration crosses a predefined threshold, the Arduino interprets this as a gas leak and immediately initiates a sequence of actions to mitigate the risk. This automatic response significantly enhances safety by removing the need for manual intervention during emergencies.

One of the main responses is activating a **servo motor** connected to the gas regulator. The servo motor is programmed to rotate and close the regulator valve as soon as a leak is detected, effectively cutting off the source of the leak and preventing further gas flow.

To alert the nearby individuals, a **buzzer** is activated, producing a loud sound to draw attention. This alarm system ensures that even those without technical knowledge can recognize that something is wrong and take necessary precautions.

In addition to the buzzer, the system includes **LED indicators** that visually communicate the system's status. A **red LED** remains ON under safe conditions, while a **green LED** lights up during a leak situation. These visual cues offer a quick and easy way to assess danger at a glance.

An **exhaust fan**, controlled via a **relay module**, is also triggered in case of a leak. This fan helps in dispersing the leaked gas and ventilating the area, reducing the concentration of flammable gases in the environment and minimizing the chances of ignition.

A **16x2 LCD display with I2C module** is used to show real-time messages and system status. This display allows users to read clear text messages like “Gas Detected” or “Safe Environment,” giving them a better understanding of the situation and increasing the system’s user-friendliness.

The entire system is powered through a stable **DC power supply**, with options for **battery backup** in case of power failure. This ensures the system’s continuous operation, especially during emergencies when safety systems are most needed.

The proposed design is compact and **modular**, making it easy to install in both new and existing setups. Users can mount it near gas sources in kitchens, storage rooms, or small industrial facilities without requiring major structural modifications.

This system is designed for **autonomous operation**, meaning that it can monitor and respond to gas leaks 24/7 without the need for human supervision. This is especially useful in scenarios where people may be asleep or away from the premises during a leak.

Maintenance of the system is simple due to the use of standardized components. All major parts—sensors, microcontroller, fan, and display—can be easily replaced or upgraded, ensuring the system's long-term usability and cost efficiency.

From a design perspective, this system promotes **low power consumption**. All electronic components selected are energy-efficient, allowing for continuous operation with minimal impact on electricity usage. This is beneficial for users concerned about utility costs.

The system's effectiveness lies in its **real-time processing and response**. There is virtually no delay between gas detection and the triggering of corrective actions. This makes it a highly reliable safety system capable of preventing catastrophic accidents.

Another advantage is its **affordability and accessibility**. Using low-cost components like the Arduino Uno, MQ-5 sensor, and servo motor ensures that the system can be implemented by households, small businesses, and academic institutions without financial strain.

This proposed solution is also **educationally valuable**, especially for engineering and technology students. It provides hands-on experience in working with sensors, actuators, embedded programming, and automation—all key elements in modern electronics and IoT systems.

Lastly, this system is **future-ready**. Though the current model does not include wireless communication features, it has been designed in a modular way that allows easy future integration of GSM or Wi-Fi modules. This ensures that the system can evolve along with technological trends.

2.3 PROJECT OBJECTIVE

The primary objective of this project is to design and develop an IoT-based Gas Leakage Detection System with an Automatic Regulator On/Off mechanism, aimed at enhancing safety and mitigating the risks associated with gas leaks. The core objective is to detect gas leaks in real-time and automatically initiate corrective actions without requiring manual intervention. This ensures the safety of residential, commercial, and small industrial environments.

A key aspect of the project is to integrate a reliable **gas detection mechanism**. Using the MQ-5 gas sensor, the system will continuously monitor the air for the presence of hazardous gases such as LPG, methane, and alcohol vapors. The sensor will provide real-time data to the Arduino microcontroller, which will then process this data to identify any potential gas leaks.

Upon detecting a gas leak, one of the first actions the system takes is to **automatically shut off the gas regulator** using a servo motor. This immediate response is crucial in preventing further gas leakage, which could potentially lead to fire or explosion hazards. This action is aimed at containing the leak and minimizing the danger posed by the escape of flammable gases.

Another important objective is to ensure **real-time alerts and notifications**. The system is equipped with a **buzzer** that activates as soon as a gas leak is detected, providing an audible alarm to anyone in the vicinity. This immediate feedback increases awareness and enables individuals to take action if they are near the leak, providing an extra layer of safety.

The system will also be equipped with a **visual display**, specifically an **LCD screen**, to show real-time information about the gas concentration and the system's operational status. This helps users understand the current situation without needing to check a separate device or app, making the system more user-friendly and convenient.

In addition to these core safety features, the system also incorporates a **ventilation mechanism** through the activation of an exhaust fan when a gas leak is detected. This fan will help clear the area of dangerous gas vapors, lowering the risk of ignition and making the environment safer for the occupants. The exhaust fan will be triggered via a relay module, ensuring that it operates as soon as the gas leak is detected.

The project also strives to make the system **scalable and adaptable** for a wide range of environments. Whether in a home kitchen, laboratory, or small industrial setup, the system is designed to be flexible and can be easily installed or expanded with additional sensors or components to cater to varying needs.

A further objective is to create an **affordable solution** that can be easily implemented without significant financial burden. By utilizing cost-effective components like the Arduino Uno, MQ-5 sensor, and basic electronic modules, the system remains affordable for homeowners and small businesses while providing high levels of protection.

The project also focuses on **ease of installation**. The system has been designed to be simple to deploy, with straightforward wiring and assembly processes. This ensures that it can be set up without the need for specialized technical knowledge, making it accessible to a broad range of users, from DIY enthusiasts to professional technicians.

Another key objective is to ensure **low power consumption** for continuous operation. Since the system is designed to operate 24/7, the components selected are chosen for their energy

efficiency, ensuring that the system can run for long periods without significantly increasing electricity costs. This is especially important for systems that will be running continuously in homes or industrial settings.

The **reliability** of the system is also a major objective. The project aims to create a system that works seamlessly under all conditions, providing consistent performance. By selecting robust components and implementing fail-safe mechanisms, the system is designed to function reliably even in challenging environments.

One of the goals of the project is to provide **long-term sustainability**. The system is built to be easily maintainable, with standardized parts that can be easily replaced or repaired. This ensures the system remains functional and cost-effective over time, making it a sustainable investment for the user.

Moreover, the system is designed to promote **safety awareness** and encourage the adoption of technology to prevent accidents. By automating the process of gas leak detection and mitigation, the system educates users about the importance of safety and promotes the use of technology in everyday life to protect against hazards.

An additional objective is to **future-proof the system**. Although the current system does not include GSM or Wi-Fi modules, it is designed with scalability in mind. Future enhancements could involve integrating wireless communication modules, allowing users to remotely monitor and control the system, ensuring that it keeps up with technological advancements.

The project also aims to provide a **practical solution** that can be easily incorporated into existing gas supply systems in both residential and industrial settings. The system's ability to integrate with existing infrastructure makes it a highly practical choice for users looking to improve safety without major modifications to their current setups.

Finally, the overarching objective of the project is to create a system that is both **innovative and educational**. Through the integration of embedded systems and IoT, the project serves as a learning tool for students and professionals interested in the fields of electronics, automation, and Internet of Things (IoT) applications. It demonstrates how modern technology can be used to address everyday safety concerns.

2.4 SYSTEM OVERVIEW

The IoT-based Gas Leakage Detection System is designed to automatically detect the presence of gas leaks and take immediate corrective actions, ensuring the safety of individuals and the environment. The system integrates various hardware components and software algorithms to provide real-time monitoring and response to gas leakage events. The system's primary function is to detect dangerous gas leaks, shut off the gas regulator, alert individuals with an alarm, and ventilate the area using an exhaust fan—all without requiring human intervention.

At the core of the system is the **Arduino Uno** microcontroller, which serves as the central unit for processing sensor data and triggering responses. It interprets inputs from the gas sensor and controls the various components based on predefined conditions. The microcontroller ensures that the system remains responsive and operational under all circumstances, making it the backbone of the entire project.

The **MQ-5 gas sensor** plays a crucial role in detecting the presence of gases like LPG, methane, and alcohol vapors. This sensor continuously monitors the air quality, sending data to the Arduino Uno. When the gas concentration exceeds a predefined threshold, the Arduino processes the data and activates the appropriate safety measures. The sensor is selected for its sensitivity, reliability, and suitability for detecting combustible gases in domestic and industrial environments.

When the gas sensor detects a leak, the **servo motor** is triggered to shut off the gas supply by controlling the gas regulator valve. This action prevents further leakage and significantly reduces the risk of fire or explosion. The servo motor ensures that the gas supply is promptly cut off, containing the leak and mitigating potential hazards associated with the gas flow.

In addition to cutting off the gas flow, the system also activates an **exhaust fan** through a relay module. The exhaust fan serves to ventilate the area and disperse the gas that has leaked into the environment. This ventilation process helps to reduce the concentration of gas in the air, lowering the chances of ignition and making the environment safer for the occupants.

The **buzzer** in the system provides an audible alarm to alert individuals nearby about the gas leak. As soon as a leak is detected, the buzzer emits a loud sound, which serves as a warning for anyone in the vicinity to take immediate action or evacuate if necessary. This ensures that even if individuals are not looking at the display, they are still aware of the danger.

The system features a **real-time LCD display** equipped with an I2C module, providing visual feedback about the status of the system. The LCD shows messages such as "Gas Detected" or "Safe Environment" based on the current gas concentration levels. This allows users to easily monitor the system's status locally and check whether any corrective action has been taken or needs to be taken.

In addition to the LCD, the system includes **two LED indicators**. The red LED lights up when the system detects a gas leak, signaling a hazardous condition. On the other hand, the green LED indicates that the system is operating normally, without any detected gas leakage. These visual indicators provide a quick, at-a-glance way for users to determine the status of the system.

The overall design of the system is focused on **simplicity** and **user-friendliness**. The hardware components are straightforward to assemble and install, ensuring that the system can be easily set up without the need for specialized expertise. The system's modular design also allows for easy maintenance, as individual components can be replaced or upgraded as necessary.

The system is designed to operate **continuously**, ensuring that gas leaks are detected and addressed 24/7. It runs on minimal power, making it energy-efficient and suitable for long-term use. This continuous operation ensures that users are always protected from potential gas leaks, whether they are at home, in a commercial setting, or in an industrial facility.

An important feature of the system is its **scalability**. While it currently serves as a standalone solution for gas leakage detection, the system can be expanded in the future to incorporate additional sensors or modules. For instance, the system can be adapted to monitor other environmental factors like temperature or humidity, further improving its versatility and applicability across various industries.

The system also provides **affordable and accessible safety**. By utilizing readily available components like the Arduino Uno, MQ-5 gas sensor, and basic actuators (servo motor, relay module, buzzer, etc.), the system remains cost-effective and within reach of a wide range of users. This makes it suitable for both domestic and small-scale industrial environments, where budgets may be limited but safety is paramount.

In terms of **reliability**, the system has been designed to withstand typical environmental conditions. The components are chosen for their durability and resilience, ensuring that they function reliably even in harsh conditions. For instance, the MQ-5 sensor is capable of withstanding environmental changes while maintaining its sensitivity and accuracy in detecting gas leaks.

The system is also designed to be **expandable**. As technology advances, additional features can be incorporated into the system, such as cloud-based data storage, remote control via mobile apps, or advanced gas sensors capable of detecting more specific or toxic gases. These enhancements would help make the system even more robust and adaptable to evolving safety requirements.

Furthermore, the system's design is **modular**, meaning that users can choose to install only the components they need, depending on the specific environment and application. For example, if the system is being used in a small home, it might only require the basic components like the gas sensor, servo motor, and exhaust fan. On the other hand, for a larger industrial setup, additional sensors and modules could be added to provide broader coverage and more advanced features.

The system also promotes **safety awareness**. By combining automation with real-time feedback and user-friendly interfaces, the project encourages individuals to take gas leak prevention seriously. The inclusion of both audible and visual indicators increases the chances of early detection and quick intervention, ultimately reducing the risk of accidents related to gas leaks.

In conclusion, the IoT-based Gas Leakage Detection System offers a practical, cost-effective, and reliable solution to a critical safety concern. By automating the gas leak detection and response process, the system ensures that gas leaks are quickly addressed, reducing the risk of fire, explosion, or health hazards. The integration of multiple safety measures, including gas detection, automatic shutoff, ventilation, and real-time alerts, creates a comprehensive solution for both residential and industrial environments.

CHAPTER 3

SYSTEM REQUIREMENTS

The gas leakage detection system requires both hardware and software components to function effectively. The essential hardware includes an **MQ-5 gas sensor** for gas detection, **Arduino Uno** as the microcontroller, a **servo motor** for controlling the gas valve, **relay module** for switching control, an **exhaust fan**, **LCD display (I2C)** for monitoring, a **buzzer** for alarms, and a **stable power supply**. On the software side, the system is programmed using the **Arduino IDE**, where logic is written in C/C++ to read sensor values and control components accordingly.

3.1 HARDWARE REQUIREMENTS



Figure 3.1 Arduino uno

The Arduino Uno is the central microcontroller in the gas leakage detection system. It processes the data from the sensors, controls the servo motor, and manages the activation of the buzzer, exhaust fan, and LEDs based on the gas detection. It acts as the brain of the system, performing necessary logic operations and decisions to ensure the safety of the environment.

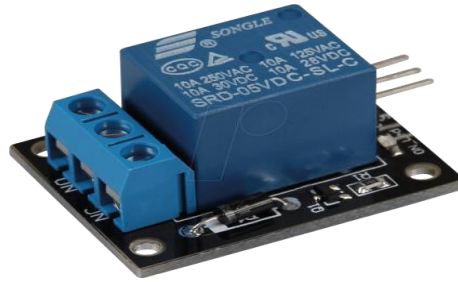


Figure 3.2 relay module

To control high-power devices, such as the water pump. The Arduino cannot power the pump directly, so a relay is used to switch the pump on and off. When the fluid level drops below a set threshold, the Arduino activates the relay, which in turn powers the pump to refill the IV bag.



Figure 3.3 relay module

The MQ-5 gas sensor is designed to detect gases such as methane, propane, and LPG. It has a sensitive material that changes its resistance when exposed to these gases. The sensor sends data to the Arduino Uno, which uses this data to determine whether there is a gas leak in the area. If a gas leak is detected, the Arduino triggers the appropriate actions to mitigate the hazard.



Figure 3.4 Servo Motor

The servo motor is used to control the gas regulator in the system. When a gas leak is detected, the Arduino signals the servo motor to either close or open the gas valve, thereby controlling the flow of gas. This automatic control helps prevent further leakage by shutting off the gas supply as soon as a leak is detected.



Figure 3.5 Buzzer

The buzzer acts as an alarm system to alert people in the vicinity of a gas leak. When the MQ-5 sensor detects a hazardous concentration of gas, the Arduino triggers the buzzer to sound an alarm, warning individuals to evacuate or take appropriate action. This audible alert ensures that the risk of gas leakage is immediately known.



Figure 3.6 Exhaust Fan

The exhaust fan is used to ventilate the area when a gas leak is detected. It helps to reduce the concentration of gas in the environment by circulating fresh air and expelling the hazardous gases outside. The exhaust fan is activated by the relay module when the Arduino detects dangerous gas levels. By improving airflow and removing potentially harmful gases from the area, the exhaust fan plays a critical role in ensuring safety during a gas leakage situation.

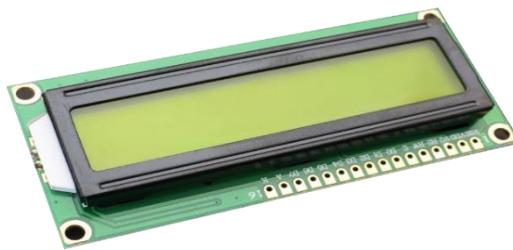


Figure 3.6 LCD Display (I2C)

The LCD with I2C module provides a real-time visual display of the system's status. It shows the gas levels detected by the MQ-5 sensor and other important information, such as whether the system is in a safe state or if a leak has been detected. The I2C module allows the display

to be connected to the Arduino using fewer pins, making it a more efficient and compact solution for output.

3.2 SOFTWARE REQUIREMENTS

Arduino Uno app, Arduino C++

Arduino Uno App

The **Arduino Uno** is a popular and widely used microcontroller board in the field of electronics and embedded systems. It is based on the **ATmega328P** microchip and is known for its simplicity, ease of use, and flexibility. The board is designed to be used with the **Arduino Integrated Development Environment (IDE)**, a software application that allows users to write and upload code to the board. The Arduino Uno provides a simple platform for developing and prototyping various electronic systems, from simple sensors to complex IoT applications.

The **Arduino IDE** is the software interface used to write and upload code to the Arduino Uno. It supports C/C++ programming languages and provides an easy-to-use interface for beginners and experts alike. The IDE allows users to write code, compile it, and upload it to the Arduino Uno via a USB cable. The Arduino Uno is then able to execute the uploaded code, making it a powerful tool for building interactive electronic projects. The IDE also supports libraries that help developers interface with various sensors, actuators, and other hardware components.

In terms of **connectivity**, the Arduino Uno uses a USB connection to communicate with the Arduino IDE on a computer. Once the code is uploaded, the board operates independently, and no further computer interaction is necessary unless additional programming or debugging is required. This makes it ideal for standalone applications, where the board can control sensors,

motors, LEDs, and other peripherals in real-time, such as in the gas leakage detection system described in your project.

Another important feature of the Arduino Uno is its real-time control capabilities. Once programmed, the board runs continuously, executing instructions and responding to sensor data without the need for manual intervention. This makes it ideal for systems that require quick reactions, such as gas leak detection, where immediate action is crucial for safety. With its real-time processing, the Arduino Uno ensures timely responses to changes in environmental conditions.

The simplicity of the Arduino Uno is paired with its powerful capabilities. It is programmed using the Arduino Integrated Development Environment (IDE), which provides an intuitive interface for writing and uploading code to the board. Users can write scripts in the Arduino programming language (a simplified version of C++) to control how the board interacts with connected components. This easy-to-learn programming environment encourages innovation and experimentation, allowing users to create custom solutions for a wide range of projects.

The **Arduino Uno app** refers to the use of mobile apps that can communicate with the Arduino Uno board. While Arduino Uno typically operates with the desktop IDE, there are also mobile applications that allow users to interact with the board through Bluetooth, Wi-Fi, or other wireless methods. These apps enable users to control devices or monitor sensors remotely using smartphones or tablets. Although the Arduino Uno board itself does not have built-in wireless communication, additional modules like **Bluetooth (HC-05)** or **Wi-Fi (ESP8266)** can be added to the board to enable mobile connectivity.

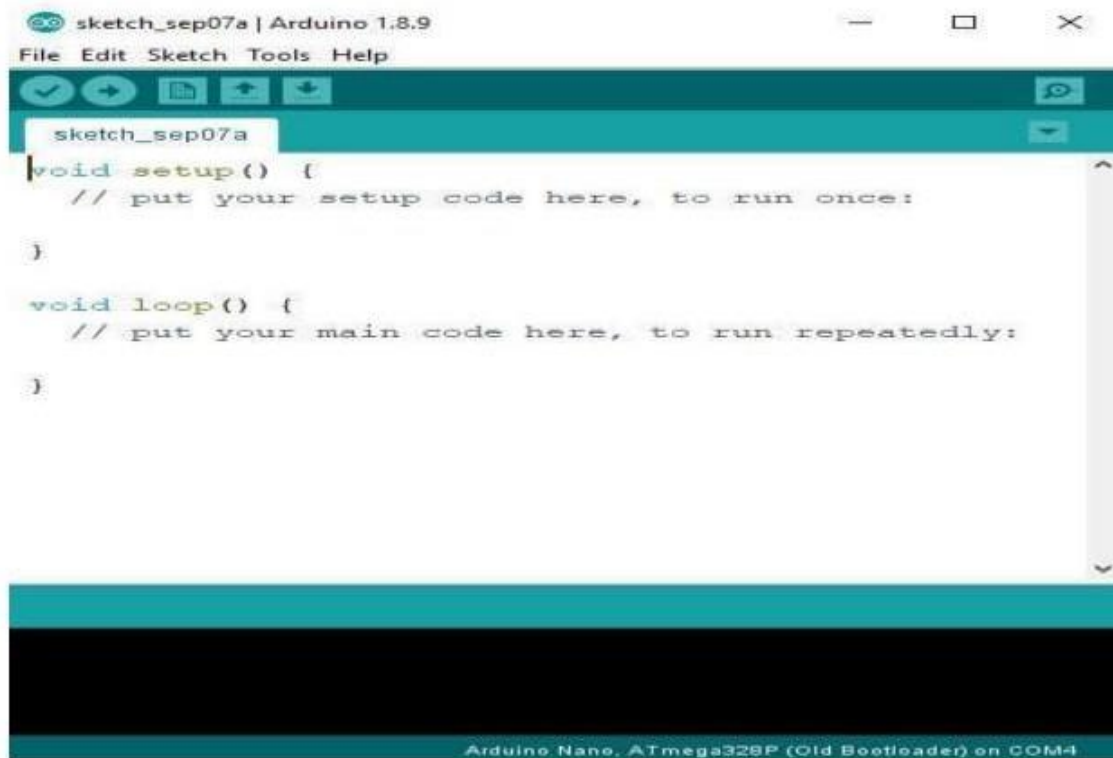


Figure 3.8 arduino app

Overall, the Arduino IDE is a must-have tool for all Arduino board development users as it simplifies the process of working and passing code to the board leader. Its user-friendly interface and clear nature make it popular with hobbyists, students and professionals

Arduino C++

Arduino C++ is a simplified version of the C++ programming language used to program Arduino boards, like the Arduino Uno, to interact with sensors, actuators, and other devices. It is a crucial part of the Arduino ecosystem, enabling users to write code that defines the behavior of the microcontroller. Arduino C++ is based on standard C++ but comes with some simplifications and modifications tailored for embedded systems development. The Arduino IDE provides an easy-to-use environment for writing, compiling, and uploading code to the board.

At the core of Arduino programming are two key functions: `setup()` and `loop()`. The `setup()` function runs once when the board is powered on or reset, and it is used to initialize settings like pin modes, communication protocols, and sensor configurations. The `loop()` function, on the other hand, runs continuously after `setup()` and contains the main code that constantly checks for inputs or triggers actions based on sensor readings. This simple structure makes it easy to understand and control the behavior of the Arduino board.

Arduino C++ supports a wide range of built-in functions, such as digital and analog input/output, timing functions, and serial communication, which makes interacting with hardware straightforward. For example, users can use functions like `digitalWrite()` and `digitalRead()` to control and read data from digital pins, while `analogRead()` and `analogWrite()` are used to handle analog inputs and outputs. This makes Arduino C++ highly versatile and ideal for controlling everything from simple LEDs to more complex systems like gas leakage detection mechanisms.

In addition to built-in functions, Arduino C++ allows users to write custom functions to organize code and make it more modular and reusable. Custom functions can be used to group specific tasks together, such as reading sensor values or activating devices, which enhances the readability and maintainability of the code. Arduino libraries, which are collections of pre-written functions for specific components, can also be included in the code to simplify working with sensors, displays, motors, and more.

One of the notable features of Arduino C++ is its accessibility to beginners and non-programmers. While C++ can be a complex language, the Arduino platform abstracts many of the complexities and provides an intuitive, approachable syntax. The use of simple functions and the Arduino IDE's straightforward interface makes it easy for users to start programming without needing advanced knowledge of C++ or embedded systems.

Arduino C++ is also compatible with various hardware components that can be connected to the board, including sensors, motors, displays, and communication modules. This compatibility allows developers to create projects that can sense the environment, process data, and respond automatically. For example, in a gas leakage detection system, Arduino C++ can be used to process data from the MQ-5 gas sensor and trigger the appropriate actions, like closing a gas valve or activating a buzzer.

The Arduino platform also encourages experimentation, allowing users to tweak their code in real-time and observe immediate changes in behavior. This makes it a great tool for prototyping and testing new ideas, especially in embedded systems where quick iterations are necessary to refine functionality. Users can start with basic projects and gradually advance to more complex systems as their understanding of Arduino C++ and embedded development grows.

Furthermore, Arduino C++ provides the flexibility to work with external hardware such as shields and sensors that add specialized capabilities to the board. Libraries designed for specific hardware can be installed and integrated into the code, making it easy to control devices like motors, displays, or communication modules. This flexibility ensures that Arduino C++ can be used for a wide variety of applications, from home automation to industrial control systems, including safety-critical applications like gas leakage detection.

CHAPTER 4

SYSTEM IMPLEMENTATION

System implementation refers to the process of integrating all the hardware components, programming the microcontroller, and ensuring that the system functions as expected. In the context of the gas leakage detection system, the implementation begins with setting up the hardware, including the Arduino Uno microcontroller, MQ-5 gas sensor, servo motor, exhaust fan, buzzer, and LCD display. The Arduino Uno serves as the central control unit that processes sensor data and triggers the appropriate actions based on predefined conditions in the software. Each component is connected to the microcontroller according to their respective pin configurations, ensuring proper functionality of the system.

The next step is the programming of the Arduino Uno using Arduino C++. The code is written to monitor the output of the MQ-5 gas sensor, which detects the presence of gases like LPG or methane. Once a gas leak is detected, the system activates the servo motor to turn off the gas valve, the buzzer to alert nearby individuals, and the exhaust fan to ventilate the area. Additionally, the system provides real-time information through the LCD display, showing status messages such as "Gas Detected" or "Safe Environment." This code is written to run continuously in a loop, with the `setup()` function initializing the system, while the `loop()` function constantly checks the gas sensor's output and takes actions when necessary.

Finally, the system is tested and fine-tuned to ensure reliable performance. The hardware and software are iteratively tested to ensure that the sensor detects gas leaks accurately, the actions (like closing the valve and turning on the fan) occur promptly, and the user receives appropriate feedback through the buzzer and LCD display. During this stage, any issues related to wiring, coding errors, or response times are identified and corrected. Once the system is fully

implemented, it offers a functional, automated solution to detect and respond to gas leaks, enhancing safety in both residential and industrial environments.

4.1 CIRCUIT DAIGRAM

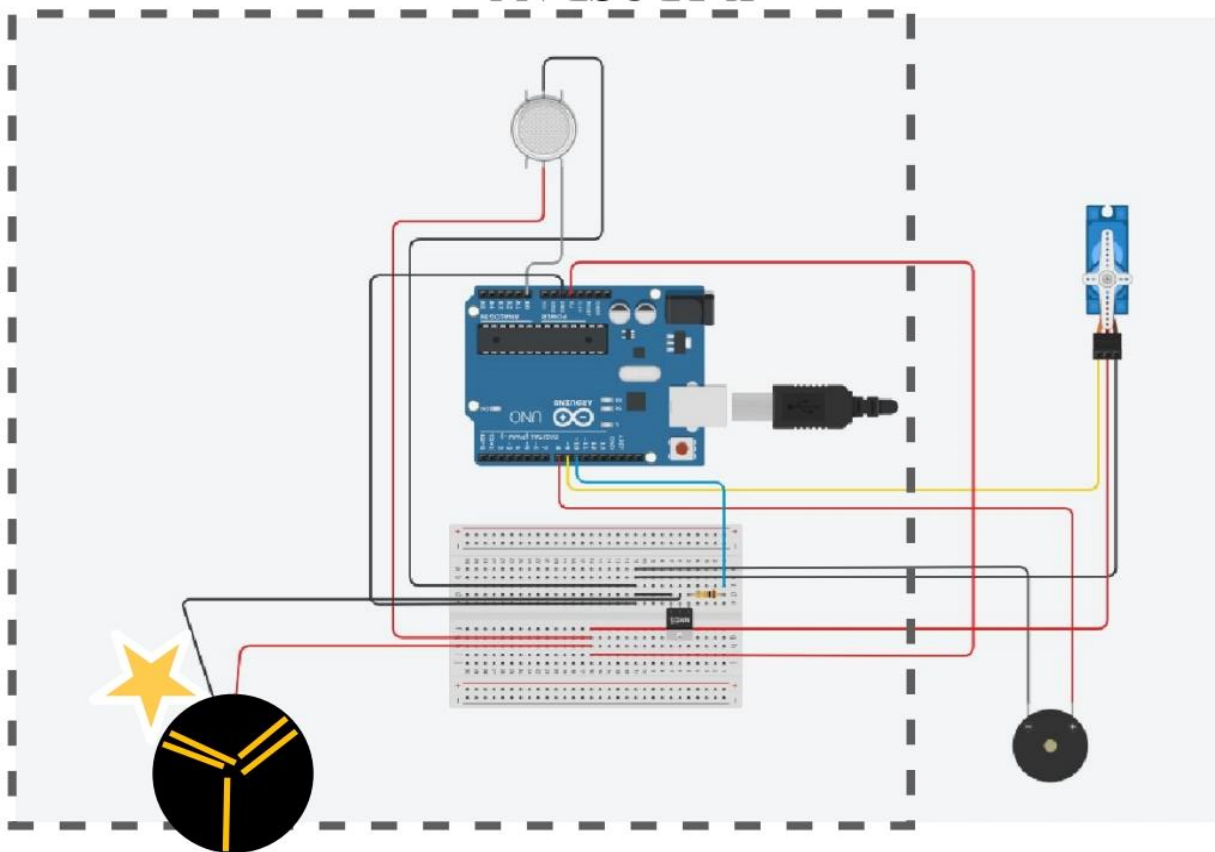


Figure 4.1 circuit diagram

The circuit diagram for the IoT-based gas leakage detection system with automatic regulator on/off mechanism involves several key components that work together to ensure the safety of a given environment. These components include the Arduino Uno microcontroller, MQ-5 gas sensor, servo motor, LCD display, buzzer, exhaust fan, and LEDs. Below is an explanation of the components and their connections.

The Arduino Uno is the central controller for the system. It processes inputs from the MQ-5 gas sensor and sends signals to control the servo motor, relay module, LCD display, buzzer, and LEDs. It operates by executing the code that is uploaded to it through the Arduino IDE. The microcontroller communicates with all other components to carry out the necessary actions when a gas leak is detected.

The MQ-5 gas sensor is used to detect the presence of LPG or other combustible gases in the surrounding environment. The sensor outputs an analog voltage that increases when gas concentrations rise, and this signal is fed to the Arduino Uno. The Arduino reads the sensor's output and determines whether the gas level exceeds a predefined threshold. If it does, the Arduino triggers the safety mechanisms.

The servo motor is connected to the gas regulator and is used to automatically shut off the gas supply in case of a gas leak. When the Arduino receives the signal from the gas sensor indicating the presence of gas, it sends a pulse signal to the servo motor to turn it and close the regulator valve. This helps prevent the further flow of gas, significantly reducing the risk of fire or explosion.

The relay module is connected to the exhaust fan, and it allows the Arduino to control the fan. When a gas leak is detected, the Arduino sends a signal to the relay to activate the exhaust fan. This helps ventilate the area by pushing out the leaked gas, reducing the concentration of dangerous gases in the environment. This action is critical to ensuring that the area becomes safe by dispersing the gas quickly.

The LCD display, along with the I2C module, shows real-time information about the system's status. It provides feedback to the user, displaying messages such as "Gas Detected" or "Safe Environment." The display helps users monitor the system's operation and understand if there

is a gas leak, even if they are not immediately nearby. This provides an extra layer of awareness and communication in case of a gas leak.

The circuit includes two LED indicators—one red and one green. The red LED is used to indicate a safe environment, while the green LED lights up when a gas leak is detected. The LED lights make the system easy to monitor at a glance, especially for users who may not be looking at the LCD display. This provides a quick, visual confirmation of the system's status, ensuring users are immediately aware of any potential hazards.

The buzzer serves as an audio alert system. When the Arduino detects a gas leak, it activates the buzzer, which emits a loud sound to alert anyone in the vicinity of the potential danger. The buzzer ensures that individuals who are nearby will be able to react quickly to the leak, even if they do not notice the LED indicators or LCD display.

The entire system is powered by an appropriate external power supply, typically a 12V DC adapter or battery pack. The power supply provides the necessary voltage and current to all components, including the Arduino Uno, sensors, and actuators. The power supply is crucial for ensuring the uninterrupted operation of the system, particularly in emergency situations where the system must remain active.

The servo motor may require a separate power source if it operates at a different voltage from the Arduino Uno. In this case, a dedicated 5V or 6V power supply can be used to ensure that the servo motor receives sufficient current to operate effectively. Proper power distribution is essential for reliable operation and to prevent damage to components.

The circuit diagram also shows the necessary wiring for all components. The gas sensor is connected to an analog input pin on the Arduino, while the servo motor is connected to a PWM pin for precise control. The relay module connects to a digital output pin on the Arduino to

control the exhaust fan, and the LCD display is connected via the I2C interface to minimize wiring complexity. Proper connections ensure smooth operation and prevent potential electrical issues or short circuits.

This circuit design is relatively simple but effective in addressing the problem of gas leakage detection. By integrating all the components effectively, it provides an automated and responsive system for managing gas leak emergencies.

4.2 WORKING PRINCIPLE

The IoT-based gas leakage detection system with an automatic regulator on/off mechanism operates on a smart, integrated process that combines real-time sensing, decision-making, and automated response. The core of the system revolves around the MQ-5 gas sensor, which continuously monitors the surrounding air for combustible gases such as LPG and methane. This sensor outputs an analog signal that reflects the gas concentration in the air, which is then sent to the Arduino Uno microcontroller.

The Arduino Uno acts as the central processing unit for the system. It continuously reads the analog input from the MQ-5 sensor and compares it against a predefined threshold value coded into the system. If the gas concentration crosses this threshold, the Arduino determines that a gas leak has occurred and immediately triggers a series of actions to mitigate the potential hazard.

The first action the Arduino initiates upon detecting a leak is shutting off the gas flow. It does this by sending a signal to a servo motor connected to the gas regulator. The servo motor rotates to a specific position that closes the regulator valve, effectively stopping the flow of gas and preventing further leakage. This mechanical response is a critical part of the system's automatic preventive mechanism.

To ensure that nearby individuals are promptly alerted to the danger, the Arduino activates a buzzer, which produces a loud warning sound. This audio alert ensures that even people not directly observing the system can be notified about the gas leak, prompting them to evacuate or take necessary precautions.

Alongside the buzzer, the system turns on an exhaust fan using a relay module. The relay acts as a switch, allowing the Arduino to control the fan, which is typically powered by a higher voltage. The fan helps ventilate the area by expelling the leaked gas, thus reducing its concentration and minimizing the risk of ignition or suffocation.

To provide real-time status updates, a 16x2 LCD display with an I2C module is incorporated into the system. This display shows messages such as “Gas Detected” or “Safe Environment,” giving users visual feedback on the condition of the environment. The I2C module simplifies wiring and reduces the number of Arduino pins needed, enhancing the overall circuit design.

The system also uses two LED indicators to give a quick visual status update. The red LED stays on under normal, safe conditions, while the green LED lights up when gas is detected. These LEDs act as intuitive, easy-to-understand visual cues for users, especially in noisy environments where the buzzer might not be easily heard.

The Arduino code is written in Arduino C++ and uploaded through the Arduino IDE. The logic in the code includes defining the sensor threshold, managing servo rotation angles, controlling the buzzer and fan through digital pins, and updating the LCD and LEDs. The code allows the system to operate automatically, reliably, and continuously.

The system components were chosen not only for their functionality but also for energy efficiency. Since the device is intended for 24/7 operation, low-power components like the

Arduino Uno and energy-efficient sensors and actuators ensure minimal electricity usage while maintaining high performance.

In essence, this system provides a complete solution for gas leak detection and response. By integrating sensing, alerting, mechanical action, and ventilation into a single autonomous unit, it delivers both safety and convenience. The simplicity of its design, combined with the effectiveness of its operation, makes it suitable for homes, labs, and small industries where gas safety is paramount.

CHAPTER 5

SYSTEM TESTING

System testing for the gas leakage detection setup was conducted to ensure that all hardware components and programmed logic functioned correctly under real-world conditions. The MQ-5 gas sensor was tested by exposing it to small amounts of LPG to verify its sensitivity and response time. The sensor successfully detected the gas and sent the corresponding signal to the Arduino Uno. Upon detection, the servo motor responded by rotating to close the gas valve, while the relay module activated the exhaust fan to ventilate the area. The buzzer also sounded an alert, and the LCD display accurately showed the gas detection message and system status, confirming the proper interaction between components.

To verify reliability, the system was tested under various environmental conditions, including different gas concentrations and repeated leak simulations. The system consistently performed the intended actions—detecting the leak, closing the valve, activating the exhaust fan, and sounding the buzzer—without delay or malfunction. These tests ensured that the system is not only responsive but also stable and dependable during continuous operation. The successful completion of system testing confirmed that the project is ready for deployment in real-world safety applications.

5.1 UNIT TESTING

Unit testing plays a crucial role in ensuring that each individual component of the IoT-based gas leakage detection system performs its function correctly before the system is fully integrated. By isolating each unit, such as the gas sensor, servo motor, buzzer, exhaust fan, and display module, developers can detect and resolve issues early in the development cycle. This

increases the reliability of the final system and reduces the chances of unexpected behavior after deployment.

The first unit tested is the MQ-5 gas sensor. The testing process involves exposing the sensor to different levels of gas concentration and verifying that the analog output values change accordingly. This helps ensure that the sensor is responsive and accurate in detecting gas presence. It is also important to confirm that the threshold set in the Arduino code correctly triggers the "leak detected" condition only when appropriate.

Next, the servo motor connected to the gas regulator is tested independently. During unit testing, specific angles are sent to the motor using the Arduino, and its physical movement is observed. The goal is to ensure that the servo rotates to the exact angle needed to close or open the regulator without hesitation, jittering, or failure. Any mechanical resistance or lag can be identified and corrected during this phase.

The buzzer is tested to ensure that it activates immediately and emits a clear, loud sound upon receiving a digital HIGH signal from the Arduino. This is essential for alerting users in real-time. Additionally, the timing and duration of the buzzer operation can be tested to ensure that it aligns with the intended warning behavior.

The relay module and exhaust fan are tested as a combined unit. The Arduino sends a HIGH signal to the relay, which should activate the fan. Testing this ensures that the relay can handle the load and that the fan starts promptly to ventilate the area. The fan's airflow and response speed are also observed to confirm it is suitable for the environment.

The LCD display with I2C module is tested to verify correct initialization, message display, and screen updates. The display must clearly show real-time messages like "Gas Detected" or

“Safe Environment.” Any issues such as garbled text or backlight failure can be caught early during unit testing.

The LED indicators are tested next. The red LED is programmed to stay ON during safe conditions, and the green LED should activate when gas is detected. Unit testing involves triggering these conditions manually through code and ensuring the correct LED response is observed consistently.

The Arduino Uno board itself is a critical unit and undergoes functional testing to verify that it can upload and execute code, communicate with all peripherals, and respond to sensor inputs as expected. This test ensures the microcontroller is fully functional and ready for deployment.

Power supply testing is conducted to ensure stable and sufficient power is delivered to all components. Voltage levels are measured across the board to confirm that the system operates without brownouts, and a backup source (if included) is also verified for automatic switching and proper functioning.

Finally, software logic testing is performed by running individual test scripts for each module. For instance, conditional loops, threshold comparisons, digital/analog reads and writes are all tested to ensure that the logic flow in the Arduino sketch aligns with the intended system behavior.

By conducting thorough unit testing for each component, the foundation is laid for a robust and dependable gas leakage detection system. This process not only improves the quality of the final product but also reduces debugging time during the integration and deployment stages, leading to a smoother project implementation.

5.2 INTEGRATION TESTING

Integration testing is a vital phase in the development of the IoT-based gas leakage detection system, as it verifies that the various hardware and software components work together as expected. Unlike unit testing, which isolates individual components, integration testing focuses on the interaction between modules, ensuring that the overall system performs seamlessly under real-world conditions. This step helps identify communication or functional gaps between components that may not appear in isolated tests.

The first phase of integration testing involves connecting the MQ-5 gas sensor to the Arduino Uno and verifying that the sensor's data is being correctly read and interpreted by the microcontroller. This test ensures that the analog signals from the sensor are being translated into actionable values and that the Arduino can detect when gas concentrations cross the predefined threshold. If the sensor and Arduino fail to interact correctly, it could lead to missed leak detections or false alarms.

Once the sensor-Arduino connection is verified, the next test involves integrating the servo motor. The Arduino is programmed to rotate the motor to shut off the gas regulator when a gas leak is detected. This step confirms whether the sensor's data triggers the appropriate motor response. The movement of the servo must be accurate and timely, as any lag or failure could result in continued gas leakage and compromise safety.

The buzzer is then connected to the system to test its response during a leak event. Integration testing checks if the buzzer activates immediately after the gas level crosses the danger threshold. This ensures that users are audibly alerted without delay. The timing, volume, and duration of the buzzer sound are verified to match the expected behavior coded in the Arduino program.

The exhaust fan, controlled via a relay module, is tested alongside the buzzer and servo motor. Once a gas leak is detected, the fan should start automatically to ventilate the environment. The relay's operation, power delivery to the fan, and fan's response time are all assessed to ensure seamless functionality. This test also checks whether the relay module correctly isolates and controls the fan's high-power requirements from the low-voltage Arduino.

The LCD display and LED indicators are tested in combination with the rest of the system. The display must show the correct messages like "Gas Detected" or "Safe Environment" in sync with the current system status. The red LED should stay on during normal conditions, while the green LED must light up when a leak is detected. Integration testing ensures these outputs change dynamically with sensor input and reflect the system state accurately.

Power management is another critical aspect of integration testing. It ensures all components receive adequate voltage and current under load. During a leak event, multiple components activate simultaneously—servo motor, buzzer, fan, LEDs, and LCD display—which can strain the power supply. Integration testing verifies that the system remains stable under such conditions and does not reset or malfunction due to power issues.

Software integration is tested to ensure that all conditional statements, loops, and delay functions in the Arduino code operate as intended when the entire system is running. The flow of control, sensor polling rate, component response time, and recovery behavior are observed. This ensures the system doesn't enter unintended states and can return to normal after handling a leak incident.

Integration testing also covers the reset and recovery process. After a leak has been detected and addressed, the system should reset to its normal monitoring mode without requiring manual

intervention. This test ensures that all indicators, outputs, and sensors return to baseline conditions, and the LCD updates accordingly.

The final phase of integration testing includes running the full system in a real or simulated environment where gas is released at a controlled rate. This holistic test verifies that all components—hardware and software—work together in a time-sensitive situation. Observations from this test help fine-tune thresholds, timings, and logic to enhance system performance.

Through these comprehensive integration testing steps, developers can ensure that the gas leakage detection system is reliable, synchronized, and ready for deployment. It confirms that individual components not only function correctly but also communicate and respond together efficiently to create a cohesive and responsive safety solution.

5.3 OUTPUT

The output images of the project illustrated as follows



Figure 5.1 Arduino Uno : Microcontroller that controls the entire gas detection system

The Arduino Uno is an open-source microcontroller board based on the ATmega328P chip. It is widely used in embedded systems and electronics projects due to its simplicity, ease of programming, and compatibility with various sensors and modules. In this project, the Arduino Uno serves as the central control unit, processing inputs from sensors and controlling the output devices accordingly.

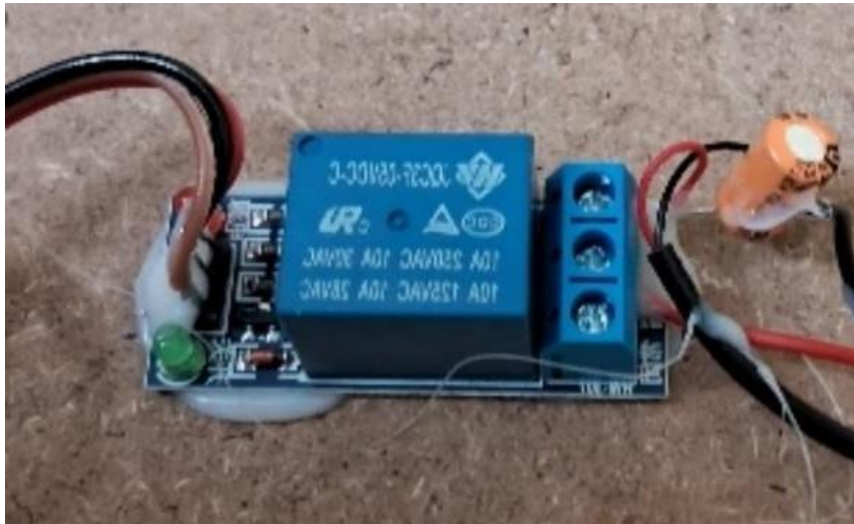


Figure 5.2 Relay Module : Switches the exhaust fan ON/OFF based on Arduino signals

Acts as an electronic switch that controls the ON/OFF operation of the exhaust fan based on the signals received from the Arduino. It ensures the fan activates automatically when a gas leak is detected, helping to ventilate the area safely.

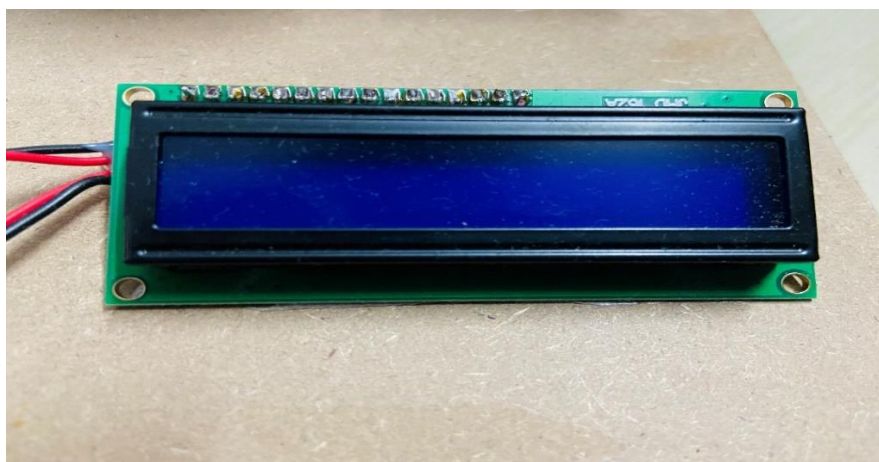


Figure 5.3 LCD Display (I2C Module) : Displays gas levels and system status messages

Shows real-time gas concentration levels and system status messages such as “Gas Detected” or “Safe Environment.” It provides clear and immediate visual feedback to the user for easy monitoring.

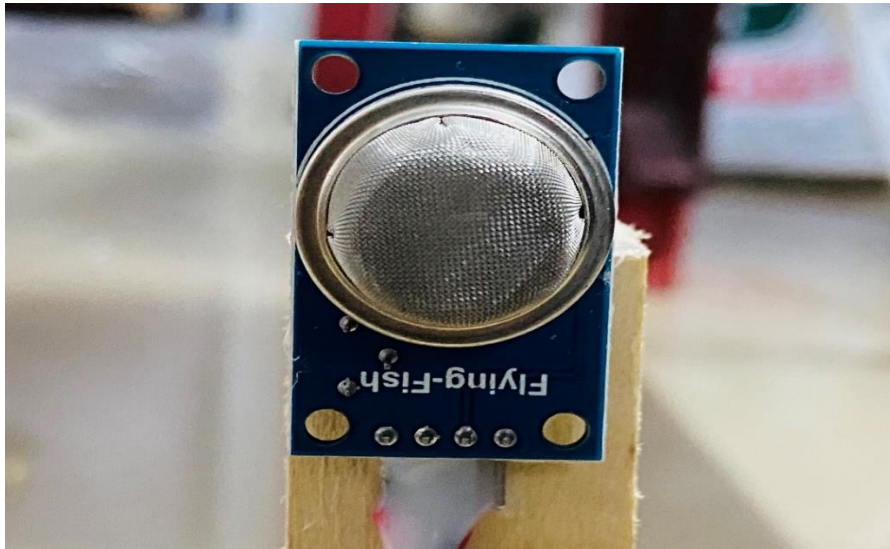


Figure 5.4 MQ-5 gas sensor : Detects the presence of LPG and methane gases.

The MQ-5 gas sensor is used to detect the presence of combustible gases such as LPG, methane, and hydrogen in the environment. It provides an analog output signal proportional to the concentration of gas detected. This sensor plays a critical role in ensuring early gas leak detection and triggering necessary safety actions.



Figure 5.5 Servo Motor : Rotates to open or close the gas valve automatically

A servo motor is a precise and compact actuator used to control the position of an object. In this project, it is employed to automatically turn the gas regulator valve on and off based on signals from the Arduino. The motor allows for accurate and reliable control of gas flow, ensuring safety during a detected gas leak.

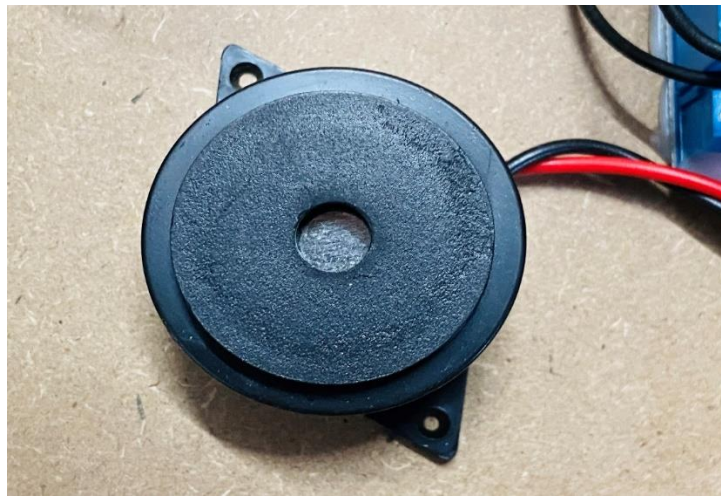


Figure 5.6 Buzzer : Sounds an alarm when a gas leak is detected

The buzzer is an output device used to generate an audible alarm when a gas leak is detected. It is activated by the Arduino to alert nearby individuals of a potential danger, ensuring that the gas leakage is noticed immediately for timely intervention.



Figure 5.7 Exhaust fan : Removes leaked gas from the room during an emergency

The exhaust fan is an essential component in the system, activated by the Arduino to ventilate the area when a gas leak is detected. It helps to disperse the accumulated gas, reducing its concentration and preventing the risk of fire or explosion. By improving airflow, the fan enhances the safety of the environment during a gas leak.



Figure 5.8 Gas Regulator : Mechanism controlled by the servo motor to stop gas flow during leakage.

The gas regulator controls the flow of gas from the cylinder to the appliance. In the event of a gas leak, the system automatically shuts off the gas flow by controlling the regulator with a servo motor. This action helps to prevent further leakage, ensuring the safety of the environment.

5.4 SAMPLE CODE

```
#include <Wire.h>

#include <LiquidCrystal_I2C.h>

// Define LCD display (16x2 with I2C)

LiquidCrystal_I2C lcd(0x27, 16, 2);


// Define pins

#define gasSensor A0 // MQ-5 Gas Sensor on A0

#define relayPin 7    // Relay for solenoid valve

#define buzzer 8      // Buzzer alarm

#define redLED 9      // Red LED (Warning)

#define greenLED 10   // Green LED (Safe)


// Gas sensor threshold

#define GAS_THRESHOLD 400


void setup() {

    pinMode(gasSensor, INPUT);
```

```
pinMode(relayPin, OUTPUT);

pinMode(buzzer, OUTPUT);

pinMode(redLED, OUTPUT);

pinMode(greenLED, OUTPUT);


digitalWrite(relayPin, HIGH); // Keep gas valve ON initially

digitalWrite(buzzer, LOW);

digitalWrite(redLED, LOW);

digitalWrite(greenLED, HIGH); // Green LED ON (Safe)


Serial.begin(9600); // Serial monitor


lcd.begin();

lcd.backlight();

lcd.setCursor(0, 0);

lcd.print("Gas Detector ON");

delay(2000);

lcd.clear();

}
```

```

void loop() {

    int gasValue = analogRead(gasSensor);

    Serial.print("Gas Sensor Value: ");

    Serial.println(gasValue);


    lcd.setCursor(0, 0);

    lcd.print("Gas Level: ");

    lcd.print(gasValue);


    if (gasValue > GAS_THRESHOLD) {

        Serial.println("Gas Leak Detected! Shutting off gas supply.");


        digitalWrite(relayPin, LOW); // Turn OFF gas valve

        digitalWrite(buzzer, HIGH); // Turn ON buzzer

        digitalWrite(redLED, HIGH); // Red LED ON

        digitalWrite(greenLED, LOW); // Green LED OFF


        lcd.setCursor(0, 1);

        lcd.print("WARNING! LEAK!!");
    }
}

```

```

        delay(5000); // Alarm duration

    }

    else {

        digitalWrite(relayPin, HIGH); // Keep gas valve ON

        digitalWrite(buzzer, LOW); // Turn OFF buzzer

        digitalWrite(redLED, LOW); // Red LED OFF

        digitalWrite(greenLED, HIGH); // Green LED ON


        lcd.setCursor(0, 1);

        lcd.print("Status: SAFE ");

    }


    delay(2000); // Wait before next reading

}

```

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENT

6.1 CONCLUTION

The IoT-based gas leakage detection system with an automatic regulator on/off mechanism provides a highly efficient and reliable solution to a critical safety concern in both residential and industrial environments. It aims to prevent hazardous situations caused by gas leaks, such as explosions, fires, or health issues. By using sensors, microcontrollers, and actuators, this system can detect gas leaks immediately and respond promptly, ensuring that any potential danger is mitigated before it becomes a major issue.

The system employs the MQ-5 gas sensor, which is capable of detecting LPG and other combustible gases. The MQ-5 sensor continuously monitors the air for any signs of gas leakage. As soon as a leak is detected, the system takes multiple corrective actions. This automatic and rapid response is essential to minimize the risks associated with gas leaks, ensuring that dangerous gases do not accumulate in the surrounding environment.

When the gas leak is detected, the system immediately shuts off the gas regulator using a servo motor. This prevents further gas from flowing into the area, significantly reducing the risk of fire or explosion. The servo motor's timely action ensures that the gas supply is halted quickly, and the potential threat is minimized. The automated nature of the system makes it far more efficient than relying on human intervention, especially in high-pressure or emergency situations.

Along with turning off the gas regulator, the system also activates an exhaust fan via a relay module. The exhaust fan helps to ventilate the area by expelling the leaked gas, further

decreasing the likelihood of an explosive atmosphere. Ventilation is a crucial aspect of ensuring the safety of the environment after a gas leak, as it helps to reduce the concentration of hazardous gases in the air.

The system is also equipped with both visual and audio alerts to inform users of the detected gas leak. An LCD display with an I2C module shows status messages such as “Gas Detected” or “Safe Environment,” providing real-time feedback on the current state of the system. In addition, a buzzer emits an audible sound to alert those nearby to the potential danger. These alarms ensure that users are aware of the situation, even if they are not monitoring the system actively.

The design of the system is focused on simplicity, affordability, and ease of use. It is easy to install and configure, making it suitable for both residential and small-scale industrial applications. The components used in the system are cost-effective, which makes it accessible to a wide range of users. Its simplicity ensures that users, even without technical expertise, can operate the system effectively and benefit from its safety features.

Another key aspect of the system is its scalability. While the current system does not include advanced communication modules like GSM or Wi-Fi, the design allows for future upgrades. In the future, these modules can be added to enable remote monitoring and control of the system. Users could receive real-time alerts via mobile apps or SMS, ensuring that they are always informed about the gas leak situation, even if they are not physically present.

The low power consumption of the system is another critical benefit. As the system operates continuously, the components are chosen to be energy-efficient, ensuring that the system can remain operational without significantly increasing electricity costs. This makes the system

ideal for long-term use in homes, industries, and any setting where continuous safety monitoring is required.

The system's integration of both hardware and software makes it a perfect project for demonstrating the principles of IoT and embedded systems. The Arduino Uno microcontroller serves as the central unit that processes data from the gas sensor and controls the system's actuators, such as the servo motor, exhaust fan, and buzzer. The integration of these components into a single, cohesive system highlights the practical applications of IoT in improving safety.

The system offers valuable learning opportunities for students and professionals working with embedded systems and IoT technology. By designing, building, and testing this system, individuals gain hands-on experience with sensors, microcontrollers, and actuators. This kind of practical knowledge is essential for anyone pursuing a career in IoT, automation, or safety engineering.

This project also helps raise awareness about the importance of automated safety systems. Gas leaks are often difficult to detect manually, and they can be extremely dangerous. This system provides an effective solution that can be implemented in real-life scenarios to prevent accidents and save lives. It emphasizes how technology can be applied to solve real-world problems, improving public safety and wellbeing.

The adaptability of the system is another advantage. While it is primarily designed to detect LPG gas leaks, the MQ-5 sensor can detect other gases such as methane, alcohol vapors, and hydrogen. This flexibility makes the system applicable to a wide range of environments, including chemical laboratories, industrial plants, and fuel storage facilities, where the monitoring of gases is essential to prevent accidents.

Incorporating such a system into real-world environments not only ensures safety but also provides peace of mind to users. The system's automatic nature means that users do not need to be actively involved in monitoring the environment, reducing the chances of human error. The system acts as an extra layer of protection, keeping individuals safe even when they are unaware of a potential hazard.

The system's effectiveness in detecting gas leaks, controlling gas flow, and providing real-time alerts demonstrates its reliability. Its autonomous operation means that users are protected, even if they are not physically present to take action. This makes it particularly valuable in scenarios where immediate intervention is necessary, but the user may not be available to respond in time.

As technology continues to evolve, this system can be further enhanced to incorporate additional features. Future developments, such as the integration of GSM or Wi-Fi modules, could provide users with even greater control and monitoring capabilities. These upgrades would allow users to receive instant alerts and take action remotely, making the system even more versatile and user-friendly.

In conclusion, the IoT-based gas leakage detection system with an automatic regulator on/off mechanism is a practical, cost-effective, and reliable solution for ensuring safety in environments where gas is used. By combining real-time monitoring, automated actions, and user-friendly alerts, the system offers a comprehensive solution to a pressing safety issue. With its scalable and adaptable design, this system is well-positioned for future improvements and can serve as a foundation for developing more advanced safety systems.

6.2 FUTURE ENHANCEMENT

Future enhancements for the IoT-based gas leakage detection system can significantly increase its capabilities and ensure a more comprehensive safety solution. While the existing system is highly effective at detecting gas leaks and taking preventive actions, future updates can include advanced features that will improve its functionality, remote monitoring capabilities, and integration with smart home systems.

One of the primary future enhancements involves the integration of GSM or Wi-Fi modules. These additions will enable remote access and control of the system through mobile apps or SMS alerts. With the inclusion of GSM or Wi-Fi, users will receive notifications on their phones when a gas leak is detected, ensuring that they are informed of a potential hazard even if they are not physically present at the location. This feature would be particularly useful for people who are away from their homes or businesses and need immediate updates on the status of the system.

In addition to remote notifications, the use of GSM or Wi-Fi could enable the system to send real-time data to cloud platforms or servers for further analysis. This would allow users to monitor the system's status over time, track gas levels, and observe patterns of usage or leakage. Cloud integration could provide valuable insights into system performance, detect recurring issues, and help predict when maintenance is required. It also opens the door for advanced analytics and predictive maintenance, which would enhance the longevity and reliability of the system.

Another key enhancement could be the inclusion of multiple gas sensors. While the current system uses an MQ-5 gas sensor, future versions could integrate sensors capable of detecting a wider range of gases. For example, sensors that detect carbon monoxide (CO), methane,

alcohol vapors, and hydrogen would make the system more versatile and capable of functioning in different environments. This would allow the system to be deployed not only in homes and kitchens but also in chemical laboratories, industrial plants, and fuel storage facilities, where the monitoring of various gases is crucial for safety.

To further improve the system's performance, an advanced AI-based algorithm could be incorporated. This algorithm would be designed to analyze the data from the sensors more effectively and make smarter decisions regarding system activation. For instance, the algorithm could learn to distinguish between different types of gas leaks, their intensity, and the likelihood of causing damage. This would prevent false positives and unnecessary alerts, making the system more accurate and efficient. The system could also integrate machine learning to improve over time, adapting to specific user environments and learning the usual gas levels.

Smart home integration is another enhancement that would provide a seamless user experience. By integrating the gas detection system with other smart home devices such as smart thermostats, lighting systems, and security cameras, users can control and monitor the system from a single interface. This integration could allow for more automated responses. For example, if a gas leak is detected, the system could automatically turn off the gas, activate the ventilation system, and alert the home security system to monitor for further hazards. It could also work with voice assistants like Amazon Alexa or Google Assistant, allowing users to query the status of the system with simple voice commands.

One of the improvements that could further enhance safety is the addition of a real-time GPS-based location tracking feature. This would be particularly useful for large industrial settings or commercial establishments. In the event of a gas leak, the system could identify the specific area of the building where the leak occurred and send alerts to relevant personnel with a map

of the affected location. This could improve response times and ensure that safety measures are taken promptly. For large-scale deployments, integrating this feature would enable efficient management and monitoring of gas leakages across multiple facilities or units.

The addition of a more advanced alerting system is another useful enhancement. While the current system uses a buzzer and LCD display, future versions could incorporate multiple alert types, including mobile push notifications, emails, and even automated phone calls to relevant personnel. This would ensure that in the event of a gas leak, the system would reach all necessary parties quickly, reducing the time taken to respond and minimizing the potential for damage or injury. Additionally, the alerts could be customized based on user preferences, such as the type of alert, the recipient of the alert, and the time of day.

A further improvement would be the integration of a battery backup system. Since the gas leakage detection system is intended to operate continuously, it is essential to ensure that the system remains functional during power outages or disruptions. Adding a battery backup would allow the system to remain active even when the primary power source is unavailable. This would ensure that gas leak detection and safety measures continue to operate without interruption, offering greater peace of mind for users, especially in critical areas where gas safety is paramount.

To enhance the system's adaptability to different environments, a more flexible user interface could be developed. Currently, the system's status is displayed on an LCD screen, but future versions could include a mobile app or web interface for users to monitor the system more conveniently. This interface could display real-time sensor data, system logs, and alert history, allowing users to make informed decisions based on the data. The user interface could also provide the ability to configure and customize system settings, including the sensitivity of the sensors and the actions taken in response to specific types of gas leaks.

A wireless mesh network feature could also be added to the system, allowing it to operate across larger areas with multiple nodes. This would be particularly beneficial in large buildings or industrial settings where gas detection needs to be distributed throughout the environment. Wireless mesh networks would allow multiple sensors to communicate with each other and with the central controller without the need for extensive wiring. This feature could improve the scalability of the system and make it more adaptable for various applications, particularly in larger commercial or industrial spaces.

To ensure that the system remains effective and reliable over time, it could include a self-diagnostic feature. This feature would periodically check the health of the system's components, including the gas sensors, actuators, and communication modules, and notify the user if any maintenance is required. Self-diagnostics would reduce the need for manual inspections and ensure that the system is always in optimal condition to detect gas leaks and respond appropriately. This would also increase the system's reliability, especially in environments where maintenance is difficult to perform regularly.

Finally, integrating a more robust security system into the IoT platform could help protect the system from cyber threats. As the system becomes more connected to the internet and includes remote access capabilities, security becomes a critical concern. Future versions could implement end-to-end encryption, two-factor authentication, and other security measures to ensure that the system remains safe from hacking or unauthorized access. This would be essential in maintaining the trust of users, especially when sensitive data is involved, such as gas leak history or user location information.

In conclusion, the proposed future enhancements for the IoT-based gas leakage detection system could greatly increase its versatility, usability, and effectiveness. With advancements in communication technologies, artificial intelligence, smart home integration, and security,

this system has the potential to evolve into a highly intelligent and automated solution for gas leak detection. These improvements will make the system more adaptable to various environments, improve safety, and provide users with greater peace of mind, ensuring the system remains a reliable safety tool for the long term.

REFERENCES

The following journals and websites were referred during the analysis and execution phase of the project.

1. **International Journal of Engineering Research & Technology (IJERT)** – Articles on IoT-based gas leakage systems and sensor integration.
2. **International Research Journal of Engineering and Technology (IRJET)** – Research papers related to embedded systems and gas detection mechanisms.
3. **IEEE Xplore Digital Library** – Technical papers on real-time monitoring systems and microcontroller applications.
4. **MQ Gas Sensor Datasheet – Hanwei Electronics Co. Ltd.** – Used for understanding MQ-5 sensor characteristics, calibration, and specifications.
5. **Arduino Official Website (<https://www.arduino.cc/>)** – Reference for programming Arduino Uno, pin configuration, and hardware libraries.
6. **CircuitDigest (<https://circuitdigest.com/>)** – Tutorials and circuit examples for interfacing sensors, relays, and servo motors with Arduino.
7. **Electronics Hub (<https://www.electronicshub.org/>)** – Guides on building embedded systems with LCDs, buzzers, and automation logic.
8. **GeeksforGeeks – Embedded Systems Section (<https://www.geeksforgeeks.org/embedded-systems/>)** – Understanding the basics of microcontrollers and control systems.
9. **Instructables (<https://www.instructables.com/>)** – DIY project references related to gas leakage detection using Arduino and sensors.
10. **YouTube Channels: GreatScott!, Tech at Home, and Electronic Clinic** – Practical demonstrations of gas leak detection projects using Arduino Uno.

